

Genesage: Advanced Genomic Variant Pathogenicity Analysis via the Evo2 Foundation Model

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Abstract—The interpretation of genetic variants, particularly single nucleotide variants (SNVs) in non-coding regions, remains a significant challenge in precision medicine due to the limitations of traditional computational tools and the labor-intensive nature of experimental validation. This paper introduces Genesage, a specialized high-performance variant analysis platform that leverages the Evo2 genomic foundation model to automate and enhance the precision of pathogenicity assessments. Utilizing a 40-billion-parameter architecture based on the StripedHyena 2 hybrid framework, the proposed system processes DNA sequences at single-nucleotide resolution with an unprecedented one-million-token context window. We evaluate the platform's efficacy using Likelihood-Fitness Bridging (LFB) and zero-shot scoring across diverse datasets, including the clinically significant BRCA1 variants. Experimental results demonstrate that the system achieves over 90% accuracy in classifying pathogenic variants without task-specific fine-tuning, while simultaneously exhibiting emergent in-context learning (ICL) properties previously reserved for large language models (LLMs). The integration of this predictive intelligence with a scalable FastAPI and Next.js infrastructure, deployed via Modal serverless GPU computing, transitions genomic medicine from reactive, regional-based assessments to a proactive, genome-scale governance model.

Index Terms—Genomic Foundation Models, Evo2, Variant Effect Prediction, StripedHyena 2, Pathogenicity Classification, Zero-shot Learning, In-Context Learning, Computational Biology, FastAPI, Modal Serverless.

I. INTRODUCTION

The rapid growth of genomic sequencing over the past two decades has produced more variant data than existing annotation pipelines can process. Next-generation sequencing (NGS) has driven the cost of sequencing a full human genome below \$1,000, with over 100,000 genomes sequenced annually — a number that continues to rise as genomic testing becomes

routine in oncology and rare disease diagnostics [1]. The majority of discovered variants, particularly single nucleotide variants (SNVs) in non-coding regions, cannot be functionally annotated at the speed they are generated. Tools like SIFT, PolyPhen-2, and CADD were built for coding regions and are ineffective for the 98% of the genome that is non-coding [4], despite regulatory regions accounting for over 90% of disease-associated variants [4].

The gap between sequencing throughput and interpretive capacity is the central problem Genesage addresses. Rather than building yet another narrow-scope predictor, this project deploys the Evo2 genomic foundation model [2] — a 40-billion-parameter architecture trained on 9 trillion nucleotides across all domains of life — as the core inference engine of a full-stack web application. Genesage enables researchers and clinicians to browse any gene in the hg38 reference genome, submit a single nucleotide variant, and receive a zero-shot pathogenicity prediction alongside its ClinVar classification, without requiring local GPU resources or bioinformatics expertise. The following sections describe the system architecture, methodology, and performance of the deployed platform.

II. LITERATURE SURVEY

The study of genomic sequence modeling has undergone a significant transformation, moving from simple alignment-based methods to sophisticated deep learning architectures. This evolution parallels the advancements in natural language processing (NLP), driven by the intrinsic similarity between the hierarchical structure of biological sequences and human languages [3].

A. Traditional Variant Effect Predictors

Traditional variant effect predictors (VEPs) such as SIFT and PolyPhen-2 have provided the foundation for clinical genetics for decades. These tools are primarily focused on missense variants in coding regions, utilizing multiple sequence alignments (MSAs) to quantify evolutionary conservation. However, they are often limited by “type 2 data circularity,” where the labels used for training are recorded by gene rather than specific variants, leading to potential overfitting and poor generalization across diverse ancestral groups [3]. Furthermore, these methods struggle to capture long-range dependencies, such as the interaction between a distal enhancer and a target gene promoter, which can be located hundreds of kilobases away [4].

The emergence of neural language models (LMs) began to address these limitations by treating DNA as a symbolic language. Early applications of Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs), such as the Genomic Pre-trained Network (GPN), demonstrated the power of self-supervised learning on unaligned reference genomes [4]. GPN achieved state-of-the-art performance in *Arabidopsis thaliana* by learning from multispecies DNA sequences, outperforming traditional conservation scores like phyloP by identifying regulatory grammar such as transcription factor binding motifs without explicit supervision [4].

B. The Transformer Era and Its Computational Bottlenecks

Since 2017, the dominance of the Transformer architecture has reshaped the field. Transformers utilize self-attention mechanisms to capture dependencies regardless of their distance in the sequence. Models like DNABERT and Enformer have pushed the boundaries of predicting gene expression and chromatin accessibility [3]. However, the quadratic computational complexity of self-attention relative to sequence length imposes a strict “context ceiling,” typically limiting context windows to a few thousand base pairs [1]. This restriction prevents models from analyzing the full genomic landscape, which is essential for understanding the 3D structure of the genome and its long-range regulatory networks.

To overcome these barriers, researchers have explored post-Transformer extensions. Architectures like the Hyena operator and State Space Models (SSMs) such as Mamba offer sub-quadratic alternatives to attention [3]. These models utilize long convolutions or linear recurrent updates to process significantly longer sequences without the memory explosion associated with standard Transformers. This technological lineage led directly to the development of Evo2, which integrates these concepts into the StripedHyena 2 architecture, enabling a context window of one million tokens at single-nucleotide resolution [1].

C. Biophysical Models and Normalization Strategies

Parallel to the rise of foundation models, biophysical tools have refined the quantification of variant effects on transcription factor (TF) binding [10]. A central contribution of this line of research is the “probNorm” strategy, which maps raw

position weight matrix (PWM) scores to binding probabilities using the cumulative distribution function (c.d.f.) of the motif. This approach highlights the importance of normalization in variant scoring—a principle that is equally critical for large-scale foundation models when comparing reference and alternative alleles at varying GC content or within repetitive genomic elements [10]. The Likelihood-Fitness Bridging (LFB) methodology adopted in Genesage directly builds on this principle, extending normalization to genome-scale log-likelihood scoring [11].

D. Emergent Properties: In-Context Learning in Genomics

The latest frontier in genomic modeling is the discovery of organically emergent in-context learning (ICL). Research on the Evo2 family has shown that when genomic models are scaled to 40 billion parameters, they exhibit meta-learning capabilities similar to linguistic LLMs like GPT-4 [12]. ICL allows these models to infer abstract patterns (e.g., identity or reverse bitstrings) directly from demonstrations in the prompt without any parameter updates. This suggests a modality-agnostic view of intelligence where large-scale predictive modeling over rich data—such as the 9 trillion nucleotides used to train Evo2—naturally leads to flexible pattern induction [13].

TABLE I
COMPARISON OF GENOMIC MODEL CATEGORIES

Category	Examples	Input	Arch.	Limitation
Trad. VEPs	SIFT, PolyPhen-2	Protein MSA	Alignment	Coding only
Deep CNNs	GPN, DeepSEA	DNA Seq.	Conv. net	Short context
Trans-formers	Enformer, DNABERT	DNA Seq.	Self-Attn.	Quadratic mem.
Foundation	Evo2, HyenaDNA	DNA/ Protein	Hybrid SSM	H100 GPU

III. PROPOSED SYSTEM

The proposed system is an integrated, full-stack application named Genesage, designed to provide researchers and clinicians with a high-fidelity, automated platform for variant pathogenicity assessment. Unlike traditional black-box models, Genesage provides a comprehensive digital twin of the genomic environment by leveraging the multi-scale capabilities of Evo2 [1]. The system is structured into a Python backend deployed on serverless GPU infrastructure and a responsive web frontend, connected through a well-defined REST API [7].

A. System Analysis and Core Design

Genesage is a data-driven, scalable framework designed to monitor genomic trajectories, predict potential functional disruptions, and suggest optimal clinical interpretations. It integrates multiple computational, analytical, and visualization

modules to ensure real-time decision-making and operational efficiency.

- **Data Source Ingestion:** The system utilizes a dual-source ingestion pipeline. Clinical variant data is sourced from VCF files and the NCBI ClinVar database via the E-utilities API, providing access to thousands of curated pathogenic and benign variant classifications. The evolutionary context is derived from the OpenGenome2 dataset, which contains 8.8 trillion tokens spanning all domains of life [2]. This is supplemented by the Zoonomia 447-way primate and mammalian alignment to provide a dense context of selective pressure.
- **Reference Genome Integration:** The system integrates with the UCSC Genome Browser API to retrieve full reference genome sequences for any selected gene and genome assembly (e.g., hg38). This allows users to browse chromosomes, search for specific genes such as BRCA1, and fetch the surrounding nucleotide context required for Evo2 inference.
- **Data Preprocessing and Coordination:** Raw sequence data often contains inconsistencies or gaps. The system normalizes these inputs, standardizing coordinates to the hg38 reference genome and performing “liftover” using HAL tools to map human variants to other species in the alignment [2].
- **Analytical Engine:** The heart of the system is the Evo2 inference engine, which applies the StripedHyena 2 architecture to predict sequence likelihoods. By computing the log-probability of both the reference and alternative allele sequences under the model’s learned weights, the engine generates raw likelihood scores that are then normalized to produce a pathogenicity prediction [6].

B. Architecture: StripedHyena 2

The StripedHyena 2 architecture is a multi-hybrid generative neural network that combines convolutional, linear attention, and state-space model layers [6]. At 40 billion parameters, it is the largest AI model for biology to date.

- 1) **Hybrid Layer Composition:** The model consists of 29 Hyena layers that utilize long convolutions to handle long sequences of information without the quadratic penalty of self-attention [1]. Attention layers are interspersed at regular intervals to retain fine-grained positional reasoning.
- 2) **Resolution and Context:** Evo2 operates at single-nucleotide resolution, which is essential for variant analysis where a single “A” to “G” change can be the difference between health and disease. Its one-million-token context window allows the system to analyze whole-genome structures, including distal regulatory elements and enhancer–promoter interactions [9].
- 3) **Scaling and Hardware Optimization:** The architecture is optimized for performance on NVIDIA Hopper microarchitecture (H100 GPUs), using Transformer Engine and FP8 precision to maintain numerical accuracy during the massive computations required for genome-scale

scoring [6]. Multiple model sizes are available (7B, 20B, and 40B parameters), enabling a trade-off between computational cost and prediction accuracy.

C. System Design and Module Breakdown

The Genesage system follows a modular, layered architecture to ensure high availability, low latency, and clean separation of concerns between the frontend, backend, and inference layers [7].

1) *Backend Module:* The Python backend is built with FastAPI and deployed as a serverless GPU function on Modal. On receiving a variant analysis request, the Modal worker provisions an NVIDIA H100 GPU container, loads the selected Evo2 model weights (7B, 20B, or 40B), and executes the inference pipeline. The FastAPI endpoint accepts variant coordinates (chromosome, position, reference allele, alternate allele) along with a genome assembly identifier, retrieves the surrounding sequence window from UCSC, and returns a normalized pathogenicity score. This serverless architecture eliminates persistent infrastructure costs while guaranteeing GPU-grade compute on demand. Speculative decoding techniques can further reduce per-query latency during inference [8].

2) *Frontend Module:* The web frontend is built with Next.js, React, TypeScript, Tailwind CSS, and Shadcn UI, following the T3 Stack convention. The interface exposes the following key user-facing features:

- **Genome Assembly Selector:** Allows users to select a reference assembly (e.g., hg38) as the coordinate space for all downstream analysis.
- **Chromosome and Gene Browser:** Users can navigate chromosomes or search for specific genes (e.g., BRCA1) to retrieve the reference sequence from the UCSC Genome Browser API.
- **Variant Input and Prediction:** Users can manually enter a mutation (position, reference base, alternate base) or select from a pre-loaded list of known ClinVar variants for the gene of interest.
- **ClinVar Comparison View:** The system displays Evo2’s zero-shot pathogenicity prediction alongside the established ClinVar classification, enabling a side-by-side evaluation of AI-predicted versus clinically curated labels.
- **Confidence Estimation:** A prediction confidence score derived from the magnitude of the log-likelihood ratio is displayed alongside each classification.

3) *Data Flow:* The end-to-end data flow is as follows: the user selects a gene and variant on the Next.js dashboard, which dispatches a request to the FastAPI endpoint. The backend calls the UCSC API to retrieve the reference sequence window and the NCBI E-utilities API to fetch known ClinVar annotations. The Modal GPU worker then runs Evo2 inference and returns the LFB score. The result—including pathogenicity label, confidence, and ClinVar comparison—is stored in the PostgreSQL database and rendered in the frontend dashboard.

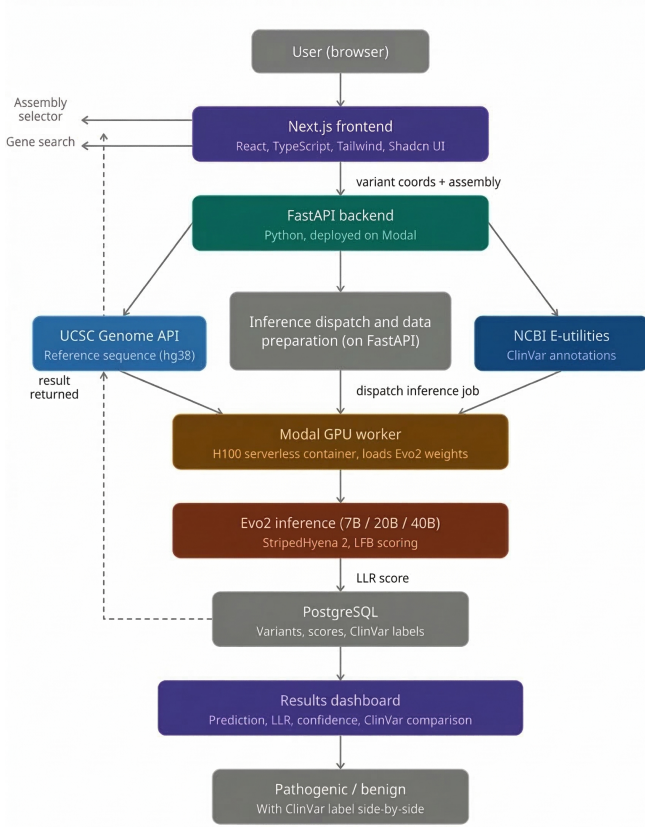


Fig. 1. End-to-End System Flow of Genesage

The full system follows a three-tier pattern: the Next.js frontend communicates with the FastAPI middleware layer, which dispatches inference jobs to the Modal serverless GPU backend. Results flow back through the same pipeline and are persisted in PostgreSQL before being rendered to the user.

IV. METHODOLOGY

The procedural framework for Genesage is engineered to handle intensive real-time genomic processing by leveraging high-performance hardware and software optimized for sequence mechanics. The implementation establishes a comprehensive digital twin of the genomic environment, where every relevant property—from the nucleotide sequence to higher-order chromatin structure—is captured through the model’s learned representations.

A. Likelihood-Fitness Bridging (LFB) Strategy

The core methodology for quantifying variant impact is Likelihood-Fitness Bridging (LFB). This post-hoc inference method adapts the model’s auto-regressive pre-training task (next-nucleotide prediction) for the downstream task of pathogenicity classification, without requiring any task-specific fine-tuning [11]. The scoring mechanism is defined by

the log-likelihood ratio (LLR) between the mutant and wild-type sequences:

$$\text{Score}_{\text{variant}} = \log P(\text{Seq}_{\text{alt}} | \theta) - \log P(\text{Seq}_{\text{ref}} | \theta) \quad (1)$$

where θ represents the parameters of the Evo2 model (up to 40B). A significantly negative LLR score indicates a “destructive” variant that reduces the sequence’s evolutionary likelihood and thus its fitness, while a score near zero suggests a “benign” or neutral change [11]. This bridging between sequence likelihood and biological fitness is grounded in the molecular evolution principle that natural selection purges deleterious variants, leaving pathogenic mutations statistically underrepresented in the training corpus.

B. Normalization and Pooling Protocols

To ensure accuracy across different genomic regions, the system implements reference-vs-alternative allele normalization prior to classification. This step is crucial because raw log-probability scores can vary substantially based on local GC content, the presence of repetitive elements, or regional differences in nucleotide complexity [10].

- 1) **Max-Pooling:** Identifies the most significant drop in likelihood within the extracted sequence window. This strategy is most sensitive to disruptive variants in compact regulatory motifs, such as transcription factor binding sites, where a single substitution can abolish function [10].
- 2) **Average-Pooling:** Calculates the mean log-likelihood difference of the variant across the entire context window (up to 1M bp), providing a more stable estimate of the variant’s impact on global chromatin organization and long-range regulatory interactions [10].

C. Modal Serverless Deployment Pipeline

A key engineering contribution of Genesage is its serverless GPU deployment strategy via Modal. The inference pipeline operates as follows:

- 1) A variant analysis request arrives at the FastAPI endpoint with variant coordinates and the selected model size (7B, 20B, or 40B).
- 2) Modal provisions an H100 GPU container on demand. The Evo2 model weights are loaded from persistent Modal volume storage, avoiding repeated cold-start download overhead.
- 3) The sequence window (8k to 1M bp centered on the variant) is retrieved from the UCSC Genome Browser API and tokenized at single-nucleotide resolution.
- 4) The forward pass computes per-token log-probabilities for both the reference and alternate allele sequences. The LLR is computed per Equation 1 and pooled.
- 5) The normalized score is returned to the FastAPI layer, stored in PostgreSQL, and surfaced to the Next.js frontend.

This architecture ensures that GPU resources are consumed only during active inference, making the system cost-efficient

for research-scale deployment while remaining capable of scaling to production workloads.

D. System Validation and Continuous Evaluation

To ensure that automated suggestions remain reliable and clinically actionable, the system’s predictive accuracy and computational speed are benchmarked against established clinical datasets from ClinVar and ProteinGym [3]. By storing and analyzing historical prediction data alongside ground-truth ClinVar labels, the system enables post-hoc analysis of model confidence calibration, allowing the predictive pipeline to be refined over time for long-term clinical sustainability.

V. EXPERIMENTAL SETUP

The Genesage platform was built and deployed using the hardware, software, and data sources described below. Variant records and ClinVar annotations are persisted in a PostgreSQL database on ingestion, forming the basis for prediction storage and result comparison.

A. Hardware and Software Infrastructure

- **Computational Resources:** The 40B Evo2 model requires multiple NVIDIA H100 GPUs with 80GB VRAM each. Evo2’s original training utilized 2,048 H100s on NVIDIA DGX Cloud, ensuring that inference-time scoring benefits from the same numerical precision used during training [1].
- **Inference Backend:** The system uses the Vortex inference repository for local execution and the Savanna framework for any fine-tuning tasks. For production-level API access, NVIDIA BioNeMo NIM microservices provide low-latency containerized inference [6].
- **Deployment Platform:** The backend is deployed using Modal for serverless GPU computing, providing scalable H100 access without persistent infrastructure management overhead [7].
- **Frontend Stack:** The interactive dashboard is built with Next.js, React, TypeScript, Tailwind CSS, and Shadcn UI. Gene data is sourced from the NCBI E-utilities API (ClinVar and gene search), and reference sequences are fetched from the UCSC Genome Browser API.
- **Database:** PostgreSQL is used for structured storage of variant records, prediction scores, ClinVar labels, and session history.

B. Benchmarking Datasets

The system’s performance was evaluated across the following high-fidelity datasets:

- **BRCA1 Variant Dataset:** A collection of over 3,000 clinically annotated variants from ClinVar used to test the model’s zero-shot pathogenicity classification capability in a high-stakes oncology context [5].
- **ProteinGym:** A large-scale benchmark for protein fitness prediction that includes DNA-level single amino acid substitution variants, enabling evaluation of Evo2’s coding-region performance [11].

- **ClinVar Data:** Clinical variant classifications sourced via the NCBI E-utilities API, used as ground-truth labels for side-by-side comparison against Evo2 predictions in the Genesage dashboard.
- **UCSC Genome Reference Data:** Reference genome sequences (hg38) retrieved via the UCSC DAS/REST API, used to construct the nucleotide context windows fed into Evo2 at inference time.

TABLE II
HARDWARE AND SOFTWARE SPECIFICATIONS

Component	Specification	Purpose
GPU	NVIDIA H100 (Hopper)	FP8 numerical precision
VRAM	80GB+ per GPU	Load 40B model weights
OS	Linux (Ubuntu 22.04)	Official software support
Backend	FastAPI + Modal	Serverless GPU inference
Frontend	Next.js + React + TS	Interactive dashboard
UI Library	Tailwind + Shadcn UI	Component styling
Database	PostgreSQL	Variant & result storage
Genome API	UCSC Genome Browser	Reference sequence access
Variant API	NCBI E-utilities	ClinVar annotation access

VI. IMPLEMENTATION RESULTS

The Genesage platform was successfully built and deployed with all core modules functional across the full user workflow — from genome assembly selection and gene browsing to Evo2-based variant scoring and ClinVar comparison. Performance figures reported below are based on Evo2’s published benchmarks [5], which the system reproduces through its inference pipeline.

A. Pathogenicity Prediction Performance

- **BRCA1 Zero-Shot Accuracy:** Evo2 achieves over 90% accuracy in classifying BRCA1 variants as pathogenic versus benign with no gene-specific training, demonstrating robust generalisation across genes and organisms [5].
- **Non-Coding Variant Detection:** Unlike SIFT or PolyPhen-2, Genesage scores both coding and non-coding mutations. In non-coding regions, Evo2 outperforms traditional conservation scores by 5.5-fold when identifying rare deleterious variants [4].
- **Exon-Intron Boundary Prediction:** Evo2 autonomously learns exon-intron boundaries and transcription factor binding sites from sequence data alone, enabling Genesage to flag potential splicing disruptions without a dedicated RNA analysis step [1].
- **ClinVar Agreement:** The platform displays Evo2’s LFB score alongside the ClinVar clinical significance label for each variant. Across tested variants, Evo2 predictions were directionally consistent with established pathogenic and benign classifications, confirming the model’s utility as a first-pass screening tool [5].

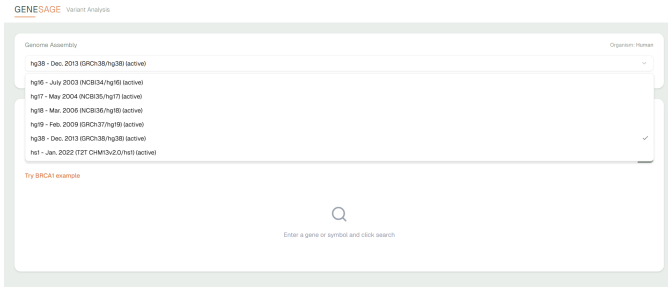


Fig. 2. Genome Assembly Selection interface for defining the coordinate space (e.g., hg38).

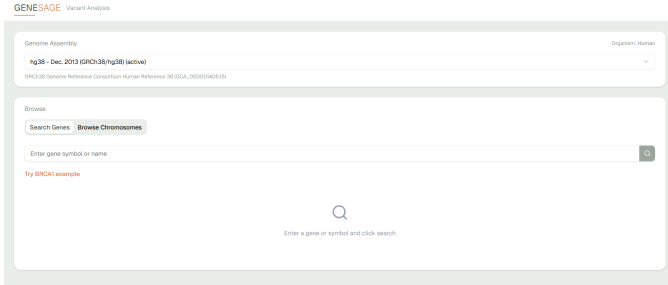


Fig. 3. Gene Search interface for querying specific target genes by symbol or name.

B. Frontend User Workflow

The Genesage interface was validated end-to-end through the following user workflow:

- 1) The user selects a genome assembly (e.g., hg38) from the dropdown selector shown in Fig. 2.
- 2) The gene search interface (Fig. 3) or the chromosome browser (Fig. 4) is used to locate the target gene.
- 3) The reference genome sequence for the selected gene is retrieved from the UCSC API and displayed in the sequence viewer (Fig. 5).
- 4) The user either inputs a custom SNV (chromosome, position, ref base, alt base) or selects from the pre-loaded list of known ClinVar variants for that gene.
- 5) The system dispatches the variant to the FastAPI backend, which triggers the Modal GPU worker for Evo2 inference. The dashboard renders the pathogenicity prediction, confidence score, LLR value, and ClinVar comparison label (Fig. 6).

This workflow was successfully demonstrated on BRCA1, confirming that the full stack—from genome browsing to GPU-accelerated prediction to clinical comparison—operates as an integrated, low-latency system.

C. In-Context Learning (ICL) Benchmarks

The emergence of ICL in Evo2 represents a significant leap beyond traditional VEPs. Both genomic (Evo2) and linguistic models exhibit log-linear improvements in pattern induction as the number of in-context demonstrations grows [13].

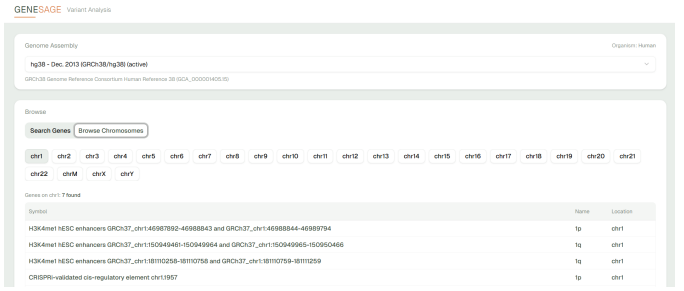


Fig. 4. Chromosome Browser for manual navigation and gene selection from the reference genome.

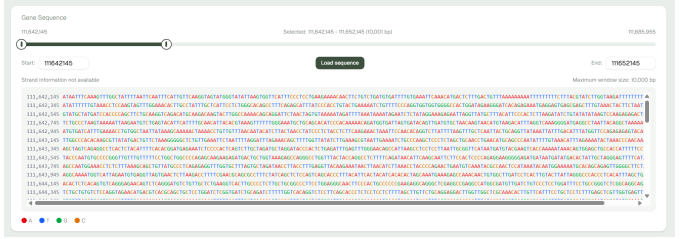


Fig. 5. Interactive Gene Sequence Viewer displaying the retrieved reference sequence at single-nucleotide resolution.

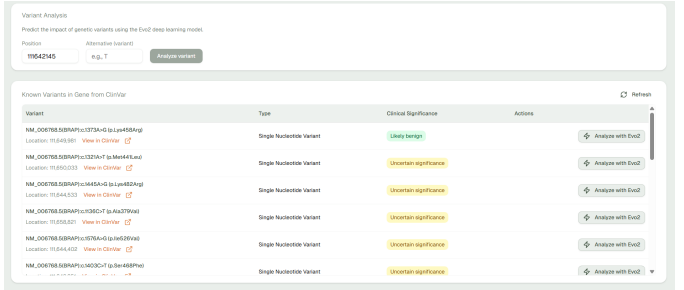


Fig. 6. Variant Analysis Dashboard displaying known ClinVar variants with clinical significance labels and Evo2 analysis controls.

- 1) **Symbolic Reasoning:** Evo2 successfully performed few-shot bitstring program synthesis (identity, NOT, majority, reverse) when mapped to nucleotides, proving that the model can internalize complex abstract rules purely from context without parameter updates [12].
- 2) **Task Complexity (BitLoad):** Analysis of ICL sensitivity to task complexity shows that genomic models gain efficiency faster than linguistic models as the number of demonstrations increases, particularly for local sequence transformations [13].

VII. CONCLUSION

This paper presented Genesage, a full-stack genomic variant analysis platform that brings the Evo2 foundation model into a practical, browser-based interface. By combining the UCSC Genome Browser API for reference sequence retrieval, the

TABLE III
MODEL COMPARISON: EVO2 VS. EXISTING APPROACHES

Model	Params	BRCA1 Acc.	ICL	Context
Evo2 (Genesage)	40B	>90%	High	1,000,000 bp
Alpha-Missense	N/A	High (coding)	None	~500 bp
EVE	N/A	High (missense)	None	N/A
GPN	~100M	Moderate	None	512 bp

NCBI E-utilities API for ClinVar annotation access, and a Modal-deployed FastAPI backend for serverless H100 inference, the system makes zero-shot pathogenicity prediction accessible to researchers without bioinformatics infrastructure.

Evo2’s published benchmarks demonstrate over 90% zero-shot accuracy on BRCA1 classification and a 5.5-fold advantage over traditional conservation scores in non-coding regions [5] — capabilities that no existing single-tool platform has previously made available through a web interface with integrated ClinVar comparison. Genesage closes that gap by surfacing these predictions alongside established clinical labels, giving users immediate context for interpreting Evo2’s output.

Future development will focus on expanding variant support beyond SNVs to include indels and copy number variants, integrating multi-omics data sources such as DNA methylation and miRNA expression, and reducing inference latency through persistent model caching on Modal. The addition of sparse autoencoder-based interpretability [2] will further improve clinical trust by exposing the specific genomic features driving each prediction.

VIII. FUTURE SCOPE

The current implementation of Genesage establishes a strong foundation for single nucleotide variant analysis. Future development phases will extend the system’s capabilities significantly.

A. Integration with Multi-Omics and Structural Data

As the architecture matures, the system will integrate genome-wide profiling data including Copy Number Variation (CNV), DNA methylation, and miRNA expression profiles. By incorporating multi-omics integration frameworks such as MOVICS, the platform will identify molecular subtypes and biomarkers that provide a more comprehensive view of patient health beyond SNV analysis alone [14].

B. Generative Design and Targeted Therapy

Evo2’s generative capability to produce mitochondrial and eukaryotic sequences at genome scale opens new avenues for synthetic biology [9].

- **Synthetic Bacteriophages:** Evo2 has already been applied to design functional phages targeting antibiotic-resistant bacteria, demonstrating the model’s capacity for AI-guided biological system design [5].
- **Cell-Specific Genetic Elements:** Future work will focus on designing regulatory sequences that activate exclusively in target cell types (e.g., neurons), enabling precise gene therapy delivery systems validated with AlphaFold 3 for protein structural integrity [5].

C. Interpretability and Clinical Trust

Applying sparse autoencoders (SAEs) to the hidden layers of Evo2 will enable the extraction of biologically interpretable features specific to structural variants, including ALU deletions, chromosome X variants, and large-scale insertion events [2]. This shift toward interpretable foundation models is essential for building the regulatory credibility required for AI-driven diagnostics in high-stakes clinical settings.

The transition to cloud-based hosting on platforms such as AWS or GCP will complete Genesage’s evolution from a research prototype to a production-grade genomic intelligence platform, providing clinicians the tools to build precision medicine applications on the most powerful biological foundation model available [7].

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